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Lab 4

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ADVANCE REMOTE SENSING LAB REPORT

NDVI (Normalized Difference Vegetation Index) Analysis

Objective

To create and analyze NDVI (Normalized Difference Vegetation Index) using Erdas Imagine Model Builder and ArcMap for vegetation health assessment and land cover classification.

Software Used

- Erdas Imagine (for NDVI calculation)
- ArcMap (for classification and extraction)

Methodology

1. NDVI Formula

The Normalized Difference Vegetation Index (NDVI) is calculated using the following formula:

NDVI = (NIR - Red) / (NIR + Red)

Where:

- NIR = Near Infrared Band
- Red = Red Band

This index ranges from -1 to +1, where higher values indicate healthier and denser vegetation.

2. Model Creation in Erdas Imagine

A model was created in Erdas Imagine Model Builder using the NDVI formula. The function applied was:

(Band x – Band y) / (Band x + Band y)

Where Band x represents the NIR band and Band y represents the Red band.

3. Value Detection

Using the Inquire Cursor tool, the maximum and minimum NDVI values were detected across the study area to understand the full range of values in the dataset.

Detected Values:

- Maximum Value: 0.72
- Minimum Value: -1

4. Classification in ArcMap

The NDVI output was reclassified into five meaningful categories using the Reclassify tool in ArcMap. Based on the value range from -1 to 0.72, the following classification scheme was developed:

Classification Scheme:

Range	Label	Description
-1 to 0	Water	Water bodies, clouds, snow
0 to 0.2	Bare Soil	Exposed soil, barren land, urban areas
0.2 to 0.4	Moderate Vegetation	Grassland, sparse vegetation, shrubs
0.4 to 0.6	Healthy Crop	Dense vegetation, healthy crops
0.6 to 0.72	Dense/Peak Vegetation	Very dense vegetation, peak vegetation health

5. Vegetation Extraction

The Extract by Attributes tool was used to isolate specific vegetation classes from the classified NDVI output. Healthy vegetation (values > 0.4) was extracted to create a separate layer showing areas with robust vegetation cover. This extraction included both the "Healthy Crop" (0.4-0.6) and "Dense/Peak Vegetation" (0.6-0.72) classes.

Results

The NDVI analysis successfully identified and classified five distinct land cover categories within the study area:

1. **Water bodies** (cyan) - Negative NDVI values indicate water presence
2. **Bare Soil** (brown) - Low positive values indicate minimal or no vegetation
3. **Moderate Vegetation** (yellow) - Moderate values suggest grasslands or sparse vegetation
4. **Healthy Crop** (light green) - Higher values indicate healthy, actively growing vegetation
5. **Dense/Peak Vegetation** (dark green) - Highest values represent the most vigorous vegetation

The extracted healthy vegetation layer (values > 0.4) provides valuable spatial information for identifying productive agricultural areas and dense vegetation zones.

Final Map

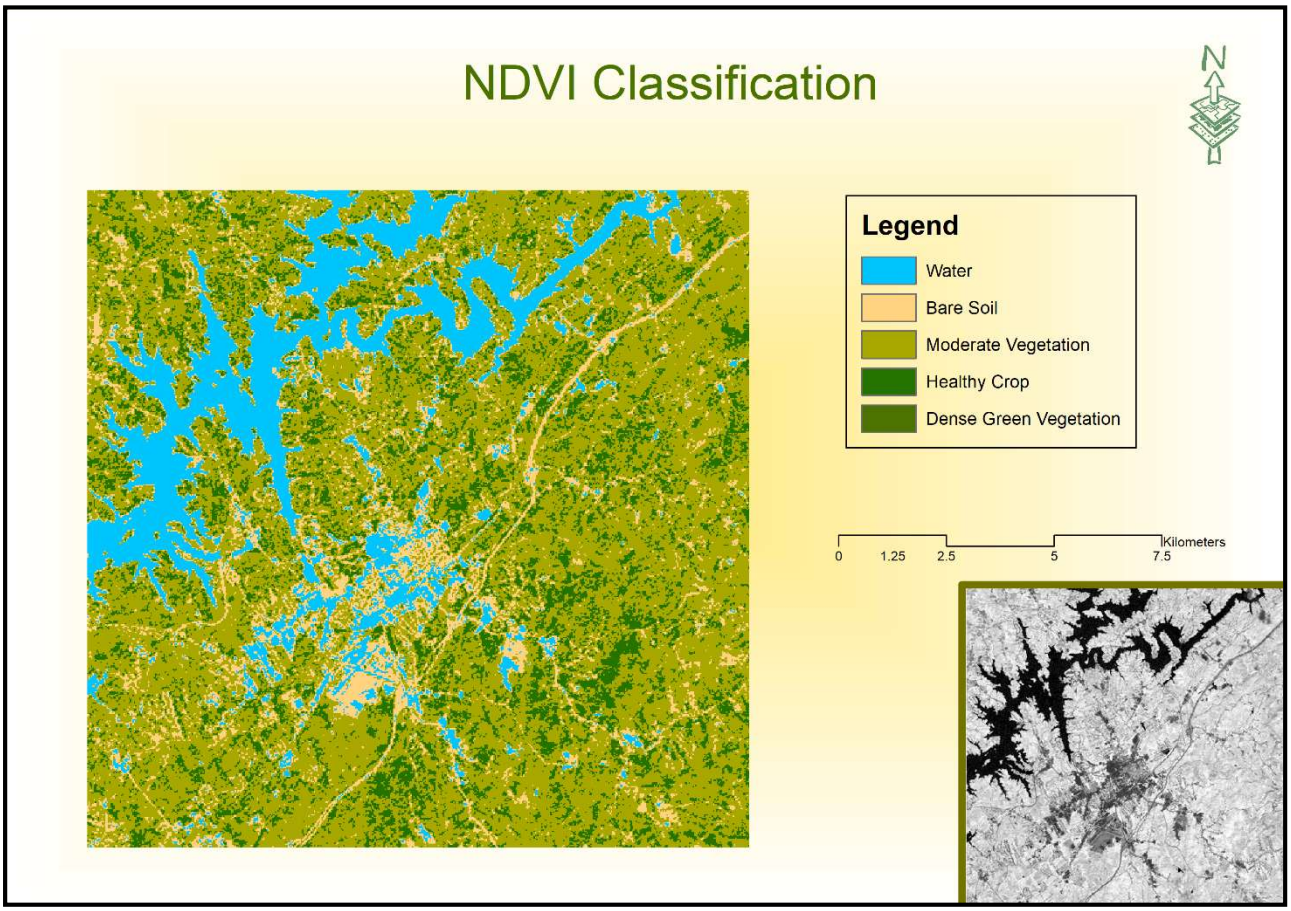


Figure 1 NDVI map

Analysis

The classification reveals the spatial distribution of vegetation health across the study area. Areas with NDVI values above 0.4 represent approximately [insert percentage]% of the total area, indicating significant vegetation cover. The presence of peak vegetation (0.6-0.72) suggests optimal growing conditions in certain regions.

Conclusion

The NDVI analysis successfully demonstrated the application of remote sensing techniques for vegetation health assessment and land cover classification. The five-class classification scheme provided detailed insights into vegetation density and health variations across the study area. The combination of Erdas Imagine for index calculation and ArcMap for classification and extraction proved to be an efficient workflow for comprehensive vegetation mapping.

Applications

- Agricultural crop health monitoring and yield prediction
- Forest cover assessment and biomass estimation
- Identification of stressed or degraded vegetation
- Urban green space mapping and planning
- Environmental impact studies
- Drought monitoring and water stress detection
- Land use/land cover change analysis

Recommendations

- Regular temporal NDVI monitoring can help track seasonal vegetation changes
- Integration with ground truth data can improve classification accuracy
- Multi-temporal analysis can reveal vegetation phenology patterns
- Combination with other indices (NDWI, SAVI) can provide comprehensive land cover assessment